

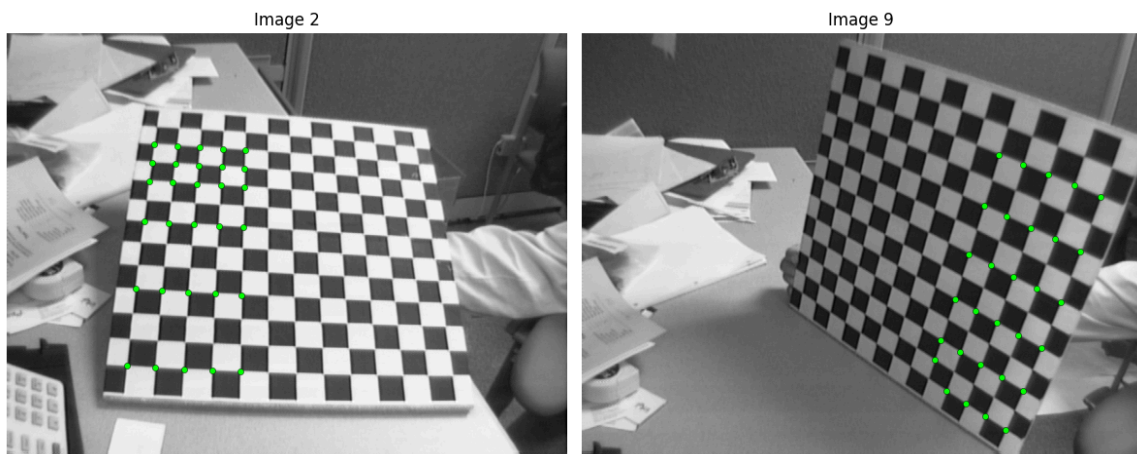
Problem 1:

Extracted a one-to-one correspondence of corner points from Image 2 and Image 9 using `findChessboardCorners()`. Then obtained the homography matrix using the Direct Linear Transform (DLT) algorithm.

Homography matrix:

```
Homography H (image 2 -> image 9):  
[[-2.26900384e+00  5.06187321e+00  9.33200306e+02]  
 [-9.29268948e-01 -1.15354922e+00  1.34865019e+03]  
 [ 1.70613214e-03  8.29184924e-03  1.00000000e+00]]  
Corner reprojection RMSE: 9.0721 px
```

Correspondences:



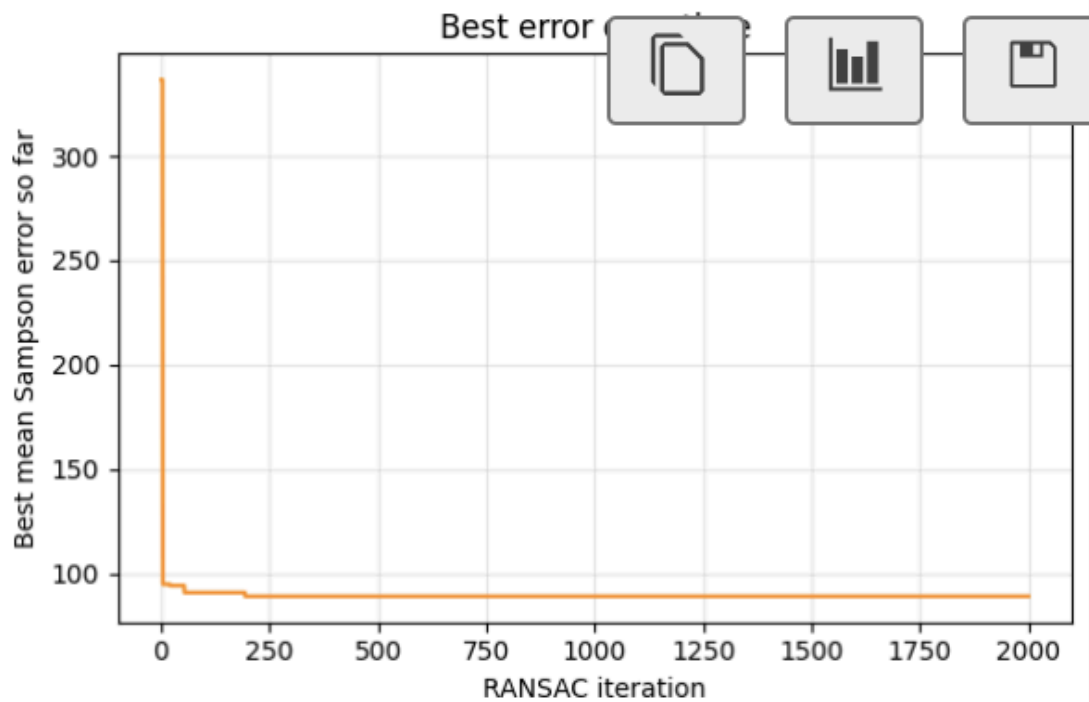
Problem 2:

Estimated good points to track unlike Problem 1 where they were defined(corners) and then obtained features for comparison using ResNet50.

Among 2000 iterations of the RANSAC algorithm, obtained the best fundamental matrix F:

```
.. Matched points: 48
Best fundamental matrix F:
[[-9.08511326e-06  1.00310634e-03 -1.66663073e-01]
 [-9.63109795e-04 -1.06204141e-05  2.02329772e-01]
 [ 1.76859614e-01 -2.24650099e-01  9.21703401e-01]]
Mean Sampson error: 88.948381
```

Best Sampson error over iterations:



Problem 3:

Video captured on campus: [campus_video.mp4](#)

Output video (original vs reconstructed): [gt_vs_3dgs.mp4](#)

Configuration:

Frame: 363 @ 30 fps

Training Iterations: 30000

Number of Gaussians: 572195

Peak Signal to Noise Ratio(PSNR): 23.483

Structural Similarity Index (SSIM): 0.695

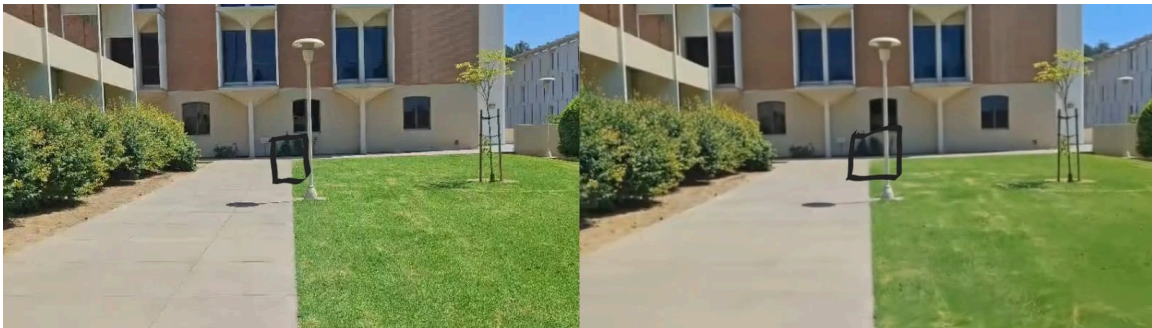
A frame from the reconstructed video:



Failure modes:

The 3DGS render was able to capture the geometry, structural depth, and lighting of the courtyard. But, we observe

1. The reconstructed video is a lot blurrier than the original, specially wrt the high frequency features like the grass and the road grooves.
2. Hazy semi-transparent floaters floating in free space where the model struggled to define sharp boundaries.



3. Geometry distortion/collapse in the white building.

